

Chapter 26. Mechanical Ventilation during Resuscitation

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Figure 26-1

Mechanical Ventilation during Resuscitation: Introduction

Before the arrival of an emergency medical service unit, ventilation given by bystanders must employ techniques that do not require special equipment. Safar, Elam, and Ruben first showed that obstruction of the upper airway by the tongue and soft palate occurs commonly in victims who lose consciousness or muscle tone and that ventilation with manual techniques is markedly reduced or prevented by such obstruction.¹⁻³ Subsequently, Safar et al^{4,5} developed techniques that prevent obstruction by extending the neck and jaw and applied these in conjunction with mouth-to-mouth ventilation. The “gold standard” today for airway maintenance during resuscitation is intubation of the trachea, which provides a route for ventilation with [oxygen](#), allows suctioning of the upper airway, protects the airway from aspiration of gastric contents, and prevents inflation of the stomach.

Spectrum of Emergency Care Skills

Airway management should be mastered by all properly trained prehospital personnel. This involves both airway assessment and airway control. Compromise of the airway may occur suddenly or slowly and progress over time; therefore, continuous assessment of the airway is vitally important. Pulse oximetry is helpful in identifying hypoxia, although hypoxia may be a relatively late sign of airway compromise. It is therefore important to evaluate breathing pattern, level of consciousness, and shortness of breath continuously.

When endotracheal intubation of children was added to paramedic practice, fifteen of 177 (approximately 8%) children were either intubated esophageally or dislodgment of the endotracheal tube were unrecognized; fourteen of these fifteen children subsequently died.⁶ Accordingly, invasive pediatric airway equipment was removed from emergency medical service units in Los Angeles County; instead, bag-valve-mask ventilation was recommended. A similarly alarming experience was seen in Orlando, Florida, when the esophagus was intubated in eighteen of 108 (approximately 16%) patients being managed by the emergency medical services.⁷ This suggests that the skills and experience of a rescuer performing basic and advanced airway management may determine if these maneuvers achieve effective oxygenation and carbon dioxide

elimination or result in extremely serious complications, such as severe neurologic impairment, or even death.⁸

Endotracheal intubation is the “gold standard” for providing emergency ventilation. Thus, every advanced emergency medical service provider must acquire and, especially, maintain intubation skills. Such a goal may be difficult to guarantee because of the large numbers of individuals who require training and/or the infrequent performance of intubations. This experience is similar to observations of the emergency medical services in Houston, Texas, where airway device-related complications were associated more with training with the devices than with the devices themselves.⁹ This confirms that the success rate of airway management interventions depends on three factors: (a) initial training, (b) continuous quality assurance, and (c) actual frequency of performing the specific intervention. For example, when the actual frequency of performing endotracheal intubation is relatively low, as in the case of the Los Angeles County emergency medical services (e.g., 2584 trained individuals performing 420 actual endotracheal intubations over 33 months), it is not surprising that endotracheal intubation performed by paramedics (who were allowed to perform bag-valve-mask ventilation) did not improve survival or neurologic outcome; instead, it caused some catastrophes.⁶ In a recent study in Germany, however, trained emergency medical service physicians intubated the esophagus in approximately 7% of all out-of-hospital intubation scenarios.¹⁰ These high incidences of intubation disasters even by highly trained rescuers actually may tip the scales toward supraglottic airway devices or bag-valve-mask ventilation for emergency ventilation of patients with a respiratory or even cardiac arrest performed by emergency medical services personnel without excellent intubation skills. Thus, teaching emergency medical services personnel a limited number or even a single airway procedure may follow the axiom “the simpler, the better” and ensure adequate oxygenation and avoid airway-related catastrophes.

Distinction between Airway Protection and Assisted Ventilation

Discussion of whether health care professionals or lay bystanders should or should not perform mouth-to-mouth ventilation has emotional connotations. The question of whether satisfactory lung ventilation in an unintubated cardiac arrest patient can be achieved, however, is clearly a scientific issue. If it is possible to identify ventilation strategies that are beneficial to patients and not harmful, then resuscitation outcomes may be improved.

The distribution of ventilation volume between lungs and stomach in a patient with an unprotected airway depends on factors such as lower esophageal sphincter pressure, airway resistance, and respiratory system compliance.¹¹ Of equal importance are specifics of techniques used for basic or advanced airway support, such as head position, tidal volume, inflation flow rate, and duration, all of which determine upper airway pressure.^{12,13} The combination of these variables determines gas distribution between the lungs and the esophagus and, subsequently, the stomach. Several fundamental differences exist between respiratory mechanics in a healthy, awake adult, an anesthetized supine patient, and a victim of a cardiac arrest ([Table 26-1](#)).

Table 26-1: Rescuer and Patient Variables Affecting Respiratory Mechanics during Mouth-to-Mouth Ventilation and Bag-Valve-Mask Ventilation

Rescuer	Patient	Conscious	Anesthetized	Cardiac Arrest
Chin support	LESP (cm H ₂ O)	~20 to 25	~20	5?
Tidal volume	Crs (mL/cm H ₂ O)	~100 to 150	~50	~20 to 50
Inflation time	Raw (cm H ₂ O/L/s)	~2 to 4	~15	?
Ventilator setting				
Bag-valve-mask size				

Abbreviations: Crs, compliance of the respiratory system; LESP, lower esophageal sphincter pressure; Raw, airway resistance.

Lower esophageal sphincter pressure is the pressure that prevents regurgitation of stomach contents into the pharynx and insufflation of air into the gastrointestinal tract during ventilation. Sphincter pressure in a healthy adult is approximately 20 to 25 cm H₂O but may be lower in patients with chronic esophageal reflux disease, during induction of anesthesia, and after insertion of a laryngeal mask airway.¹⁴ It is unclear, however, whether these changes in sphincter pressure result from induction of anesthesia. Animal investigations showed that the lower esophageal sphincter pressure deteriorated rapidly from a baseline level of 20 to 5 cm H₂O within 5 minutes of untreated cardiac arrest.^{15,16} This decrease in lower esophageal sphincter pressure was also measured in human subjects following cardiac arrest.¹⁷ In a healthy adult, air flows freely from the lips to the alveoli, and therefore, minimal inspiratory pressure is required to move gas during spontaneous ventilation. In a patient with chronic obstructive lung disease, airflow through the respiratory system may be impaired by mucus or airway spasm.¹⁸ Data from clinical studies showed an airway resistance of 2 to 4 cm H₂O/L/s in healthy volunteers, 8 to 15 cm H₂O/L/s in patients with varying degrees of lung disease, and 17 cm H₂O/L/s with positive-pressure bag-valve-mask ventilation in a patient during induction of anesthesia.¹⁹ In a healthy awake adult, 1 cm H₂O of inspiratory pressure moves approximately 100 mL of air into the lungs; compliance decreases to 50 mL in an anesthetized supine adult and may decrease to approximately 20 to 50 mL in a cardiac arrest patient.²⁰ Many factors can change lung compliance. Pulmonary vascular congestion during cardiac arrest increases the volume of the parenchymal interstitium and reduces compliance. Chest compressions and pulmonary edema secondary to left-ventricular failure also contribute to a reduction in compliance.²¹

Chin support and backward tilt of the head are the two most important maneuvers during rescue ventilation of paralyzed and nonparalyzed patients with an unprotected airway. Moreover, a slight lateral tilt of the head may reduce upper airway obstruction arising from backward relaxation of the tongue and soft tissues.²² The settings of an automatic ventilator or the technique of bag-valve-mask ventilation governs inflation flow rate, inflation time, and tidal volume, all of which affect peak inflation pressure, assuming that there is no significant upper airway obstruction.²³ The relationship between peak inflation pressure and lower esophageal sphincter pressure determines gas distribution between the stomach and lungs.²⁴ For example, when peak inflation pressure exceeds sphincter pressure, some air will flow into the stomach; if sphincter pressure is higher than peak inflation pressure, inspiratory air will flow completely into the lungs. The preceding scenario, however, depends on the assumption that the lower esophageal sphincter pressure acts like a mechanical valve, which may not be the case. For example, Safar²⁵ stated that stomach inflation in healthy volunteers was self-limiting, but lower esophageal sphincter pressure was not reported. This observation implies that the physiology of respiratory mechanics in human subjects may be more complex than has been considered previously. Until a better understanding of these mechanisms is obtained, the goal of ventilation with an unprotected airway is to keep it permanently patent and to keep peak inflation pressure to a minimum at all times to prevent stomach inflation.

Stomach inflation is a complex problem that may cause regurgitation,²⁶ aspiration,²⁷ pneumonia,²⁸ and possibly death.²⁹ Stomach inflation will increase intragastric pressure,³⁰ elevate the diaphragm, restrict lung movements, and so reduce respiratory system compliance.³¹ A reduced respiratory compliance may direct even more ventilation volume into the stomach when the airway is unprotected,¹⁰ thereby inducing a *vicious cycle with each breath*.³² The life-threatening complication of stomach inflation, regurgitation, and subsequent aspiration pneumonia (induced by gastric acid) causes a loss of alveolocapillary integrity and pulmonary surfactant. As a result, fluid and proteins pass into the interstitial spaces, alveoli, and bronchi, causing pulmonary edema, decreased pulmonary compliance, increased lung weight, and significant intrapulmonary shunting or ventilation-perfusion mismatching. Consequently, alveolar gas exchange is impaired markedly. Data from an animal model showed that aspiration of more than 0.8 mL/kg of body weight resulted in 50% mortality. Extrapolation of these results to humans suggest a critical aspiration volume of 50 mL at a pH of 1.³³ The results from recent studies of anesthetic practice, however, suggest that the true morbidity from acid aspiration has been exaggerated greatly.³⁴

Cricoid Pressure and the Sellick Maneuver

A maneuver for preventing stomach inflation during ventilation with an unprotected airway is to apply cricoid pressure; this is a simple, effective maneuver to prevent stomach inflation³⁵ that was first described 200 years ago.³⁶ In a model of human cadavers, an intraesophageal pressure of 75 cm H₂O was required to overcome cricoid pressure, indicating that the Sellick maneuver may be able to prevent gastric distension even when ventilating with a high peak inflation pressure.³⁷ One study investigating the efficacy and

minimum inflation pressure at which gas entered the stomach in pediatric patients found that appropriately applied cricoid pressure was invariably effective in preventing gas insufflation into the stomach in all children with and without paralysis.³⁸ The value of cricoid pressure, however, has been established based on case report series and cadaver and animal studies, and has never been validated in a large, controlled prospective study. Because of the lack of prospective studies, some authors regard the level of evidence for cricoid pressure as V (expert opinion) and the level of recommendation as D (levels being A to D, with A being the highest recommendation level).³⁹ The value of cricoid pressure has been further questioned because of the risk of esophageal rupture when vomiting occurs.⁴⁰ A further technical limitation is that applying both cricoid pressure and bag-mask ventilation by a single rescuer is nearly impossible and requires a second person to assist.

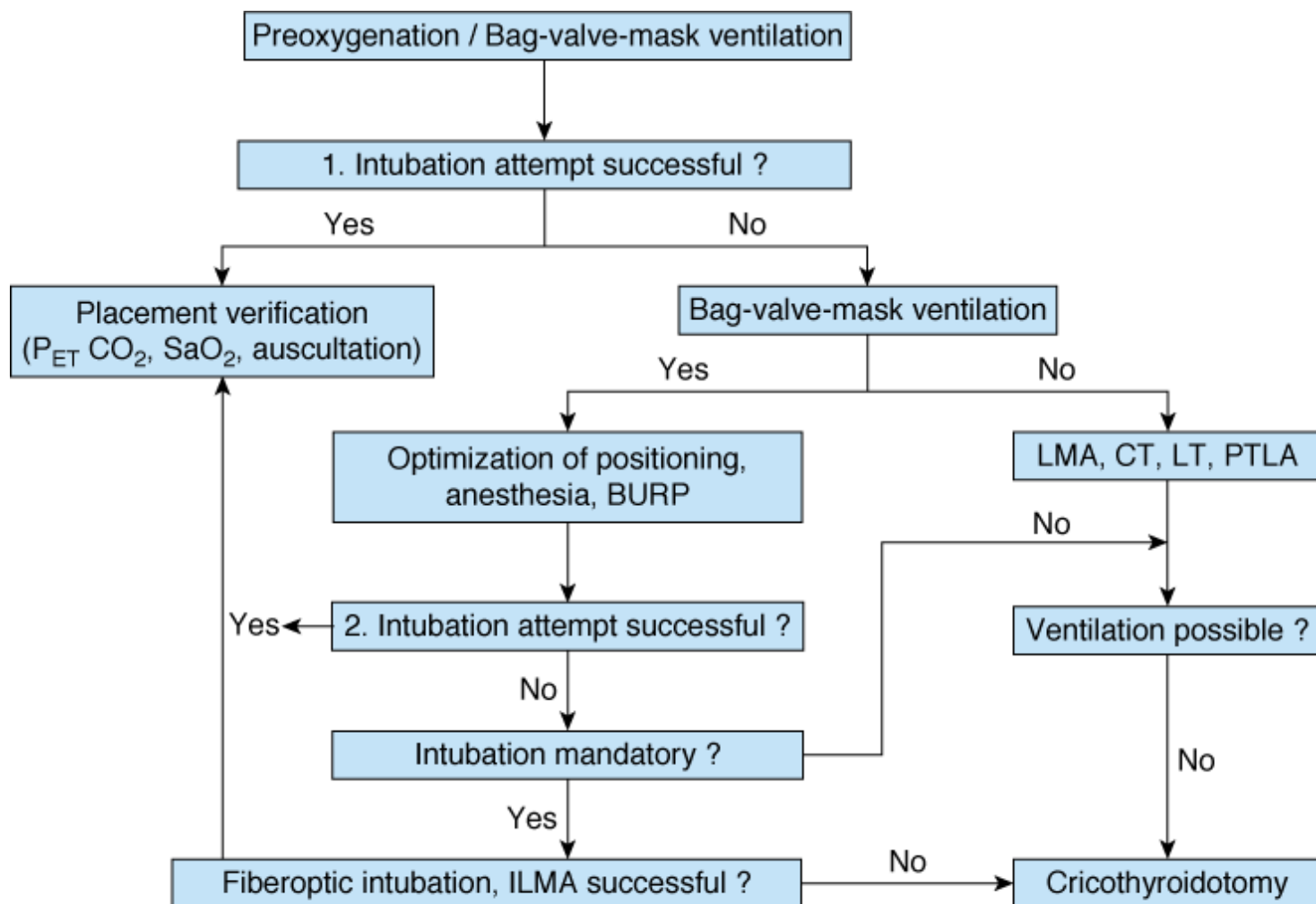
Moreover, a possible complication of cricoid pressure is airway obstruction.^{40,41} In fifty-two anesthetized patients, airway obstruction did not occur without cricoid pressure, occurred in one patient (2%) with cricoid pressure of 30 N (3 kg or approximately 6.6 lb), in twenty-nine patients (56%) with cricoid pressure of 30 N (3 kg or approximately 6.6 lb) applied in an upward and backward direction, and in eighteen (35%) patients with cricoid pressure of 44 N (4.4 kg or approximately 9.7 lb).⁴⁰ These findings were validated in a recent study in children.⁴² Furthermore, cricoid pressure deforms the larynx and impedes laryngoscopic visibility for intubation and thus may result in more intubation difficulties.⁴³ Finally, modern magnetic resonance imaging has questioned whether cricoid pressure truly closes the hypopharyngeal space or upper esophagus secondary to lateral displacement of the esophagus.⁴⁴ Thus, in regard of all these limitations it is not surprising that the European Resuscitation Council's 2010 cardiopulmonary resuscitation (CPR) guidelines stated: "The routine use of cricoid pressure in cardiac arrest is not recommended. If cricoid pressure is used during cardiac arrest, the pressure should be adjusted, relaxed or released if it impedes ventilation or intubation."⁴⁵

Role in Different Settings

Prehospital Care

Securing the airway by tracheal intubation in the prehospital setting may be very difficult. If airway management is difficult in a prehospital patient, maintenance of oxygenation can be in doubt despite use of a bag-valve-mask. Definite securing of the airway can be done later in the emergency department with more experienced personnel and equipment. In acute life-threatening situations, however, a standardized procedure, such as an algorithm, may be fast and reliable (Fig. 26-1).

Figure 26-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*,
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Algorithm for airway management. *BURP*, backward, upward, rightward pressure maneuver; *CT*, Combitube; *ILMA*, intubating laryngeal mask airway; *LMA*, laryngeal mask airway; *LT*, laryngeal tube; *PTLA*, pharyngotracheal lumen airway; *P_{ET}CO₂*, end-tidal carbon dioxide; *SaO₂*, arterial oxygen saturation.

Cardiopulmonary Resuscitation

Outcome after CPR in adults with chest compression alone was similar to that after chest compressions combined with mouth-to-mouth ventilation; Hallstrom et al. concluded that chest compressions alone may be the preferred approach for bystanders inexperienced in CPR.⁴⁶ It has been suggested that more lives could be saved if only rapid chest compressions are performed (producing immediate reperfusion of vital organs but with decreased oxygen content) in contrast to the “gold” solution of rapid chest compressions combined with ventilation, which may produce less perfusion but with higher oxygen content.

In a more recent prospective study in Japan, chest-compression-only CPR resulted in increased survival rates in adults.⁴⁷ Results, however, have been inconsistent in follow-up studies.^{48,49} In a 2011 study, there seemed to be a time-dependent factor with regards to survival rates: The longer CPR was performed by lay rescuers, the more dependent ventilation became. This is rational from a physiologic viewpoint; at the beginning of a

cardiac arrest caused by cardiac disease, there is still enough air in the lungs to enable sufficient oxygen uptake as long as chest compressions are able to deliver any circulatory output. Moreover, directly after cardiac arrest gasping may additionally ventilate the patient's lungs if the airway is in a patent state.⁵⁰ This reasoning is in accordance with a study in children where compression-only CPR seemed to be less effective. Because pediatric cardiac arrest contrasts with adult cardiac arrest, being more often caused by hypoxemia than cardiac disease, superiority of ventilation CPR is not surprising in pediatric patients.⁵¹ In a metaanalysis, authors detected a potential benefit for compression-only CPR, with a "number needed to treat" of forty-one, based on prospective studies; in contrast, analysis of retrospective studies did not reveal any benefit for chest-compression-only CPR.⁵² The 2010 CPR guidelines recommend that trained lay rescuers provide ventilation; untrained or unwilling rescuers should provide chest-compression-only CPR.⁵³ Chest compressions and ventilation are performed in a 30:2 ratio (thirty compressions to two ventilations) in a cardiac arrest when the airway is not secured.⁵³ After intubation, ventilation is performed out of synchrony with chest compressions; a respiratory rate of 10 breaths/min is recommended.

One argument that ventilation during CPR could be abandoned is that gasping provides sufficient gas exchange. The value of gasping in human subjects has to be interpreted with caution because fundamental differences in human and animal upper airway anatomy make it difficult to extrapolate results from laboratory studies, where gasping was beneficial. The human upper airway is kinked and is subject to rapid occlusion by the tongue and/or head position in the supine position, whereas the upper airway in swine, dogs, and rats is straight and may not be occluded by the tongue or head position in the supine position.⁵⁴ In addition, animal studies were performed in fasting animals that, as a consequence of fasting, are extremely unlikely to aspirate gastric contents. Furthermore, it is unknown if humans gasp as frequently or as deeply as animals during cardiac arrest. Whether gasping in humans results in effective gas exchange has not been studied, and for ethical reasons, such a study would be extremely difficult to perform. If a patient is gasping and the rescuer is unwilling or unable to perform mouth-to-mouth ventilation during basic life-support CPR, one strategy is to keep the airway open and perform chest compressions alone until emergency medical services arrive so as to provide a minimum of ventilation.⁵³ In the absence of a dedicated airway, extending the head by placing a pillow under the shoulder or gently tilting it laterally could reduce the acute pharyngeal obstruction from the tongue falling backward over the palate.

Ventilation consumes time during CPR that could be devoted to chest compressions.⁵³ Because only rapid chest compressions are effective, some argue that time spent for ventilation should be decreased and the time for chest compressions increased. Because cardiac output during CPR is at best approximately 25% of normal, some argue that the low cardiac output does not need to be accompanied with a normal minute ventilation. This argument has yet to be proven in a clinical study. Until evidence is available, assisted ventilation should be performed during CPR whenever possible, although the rate of chest compressions should not be decreased critically by ventilation attempts.

Another method, first being described in the late 1990s, is to decrease intrapulmonary pressure by intermittent airway occlusion during the chest-decompression phase. Therefore, a special impedance-

threshold valve was designed that is adjusted to a tracheal tube and occludes the airway in the decompression phase.⁵⁵ This occlusion results in decreased intrapulmonary pressure that enhances venous return to the heart and subsequently increases cardiac output with the next chest compression, improving oxygen transport to vital organs in cardiac arrest. The impedance-threshold valve is especially effective when intrapulmonary pressure is further reduced by active chest decompression using special suction cups attached to the sternum.⁵⁵ In a clinical study, use of such an impedance-threshold device significantly increased the proportion of patients that “survived to hospital discharge with favorable neurological function” (from 6% to 9%).⁵⁶ It should be noted, however, that active ventilation is mandatory using the impedance-threshold device, because continuous airway occlusion may otherwise result in atelectasis compromising pulmonary function.⁵⁷

Trauma

The underlying principle for care of trauma patients is the ABCs (*airway, breathing, circulation*) of resuscitation. Initial priorities after trauma are the same for the field and the hospital, but field providers usually have only limited equipment and limited assistance. They also provide care in the least-controlled circumstances, such as the roadside. Airway management is the highest priority and should occur as early as possible; in severe cases, it may be necessary in the field. Physical examination may reveal subcutaneous emphysema, absent breath sounds, and unilateral hyperresonance, identifying a tension pneumothorax, which can be decompressed. Progressive clouding of consciousness indicates nearly always a loss of airway protection, risking aspiration; endotracheal intubation is required. Oral intubation is the most common method because adequately sized tubes can be placed under direct vision. Nasal intubation requires significant expertise, especially when performed blind, and may exacerbate agitation; it is contraindicated in patients with midface fractures. Laryngoscopy in an awake patient nearly always worsens agitation; in these instances, rapid-sequence intubation (RSI) is advised.

RSI involves administration of a sedative followed by a short-acting muscle relaxant, typically succinylcholine. If endotracheal intubation fails, however, the patient must be ventilated with a bag-valve-mask device until short-term paralysis resolves. A retrospective analysis of trauma patients revealed that RSI in the field and in hospital were equally successful (97.9% vs. 98.5%) and safe.⁵⁸ Other studies reported success rates of 84% to 90.5% for field intubations, suggesting that training and experience are the major determinants of success.⁵⁹ Failed intubation was associated mainly with inadequate relaxation and difficult anatomy (i.e., morbid obesity or inability to visualize the vocal cords). Although the length of hospital stay and mortality were comparable, pneumonia occurred more frequently with RSI in the field than in hospital.⁶⁰ In a prospective study of severely injured trauma patients who were intubated in the field via RSI or immediately intubated on admission to the hospital, total out-of-hospital times were twice as high for the field group versus the hospital group (26 vs. 13 minutes).^{59,61} Patients intubated in the field versus in the hospital had more ventilator, hospital, and intensive care unit days, higher mortality (23% vs. 12.4%), and a 1.5-fold increased risk of nosocomial pneumonia.⁶² These data argue against spending extra time in the field with severely or even

moderately injured patients (injury severity score 15 to 25).⁶³ Even in recently published studies, conducted several years after the problems of high failure rates in out-of-hospital emergency airway management had been identified, the rate of failed intubations is still unacceptably high.⁶⁴

The high failure rate may indicate that although proper ventilation and perfect intubation skills in the field had been identified as being of utmost importance, it is not possible to achieve in all cases. Rescue personnel must recognize the importance of skill retention. Thus, before intubation attempts, a risk-to-benefit assessment is needed to prevent harm with multiple failed attempts. For example, if conditions are difficult (morbid obesity or midface fractures), it may be prudent to use a bag-valve-mask device and subsequently intubate immediately after arrival in the emergency department. Health care personnel, however, must understand that bag-valve-mask ventilation needs to be performed carefully. In laboratory shock models, excessive ventilation with a bag-valve-mask caused increasing positive intrathoracic pressure, inhibition of venous return, decreased coronary perfusion, and thereby decreased survival.^{65,66} In contrast, reduced intrapulmonary pressure resulted in better venous return in severe hemorrhagic shock and thus improved blood flow in an animal model comparable to CPR.⁶⁷ In contrast to the impedance-threshold device during CPR, however, large prospective studies determining the effect of decreased intrapulmonary pressure in shock patients are still lacking.

Cervical Spine Injury

Cervical spine injuries occur in 2% to 5% of blunt trauma patients,⁶⁸ and approximately 10% of these are unstable.⁶⁹ The most common fracture was C2, accounting for 24% of fractures. Dislocations are most common at C5-6 and C6-7.⁷⁰ Missed cervical spine injury can have disastrous consequences. The incidence of missed or delayed diagnosis is 1% to 5%, and up to 30% of these patients develop secondary neurologic damage.⁷¹ The primary objective when managing the airway of such a patient is to minimize neck movement while the airway is secured rapidly and efficiently. Flight paramedics had only a 52% success rate with blind nasal intubation in spontaneously breathing patients in an out-of-hospital setting but were successful with this procedure in fourteen of fifteen patients in the hospital.⁷² Traumatologists working in a prehospital environment used RSI with a success rate of 97% to 99%.⁷³ When emergency physicians were compared with anesthesiologists using RSI in an out-of-hospital environment, the nonanesthesiologists were twice as likely to fail to intubate and needed to undertake a surgical airway.⁷⁴

During normal direct laryngoscopy and oral intubation, significant extension occurs between the occiput and C1 and between C1 and C2.⁷⁵ In an emergency department, the most common approach to intubating the multiple-injury patient with a potential cervical spine injury is RSI while maintaining cricoid pressure and manual inline neck stabilization. Manual inline neck stabilization is provided by an assistant who holds the patient's mastoid processes firmly down, opposing the upward forces generated during laryngoscopy.⁷⁶ In anesthetized patients, this maneuver reduced head extension by 50% and should increase the safety of direct

laryngoscopy.⁷⁷ Axial traction must be avoided because excessive distraction may injure the spinal cord. The view at laryngoscopy is better with manual inline neck stabilization than with rigid-collar immobilization⁷⁸; if a cervical collar is left on during laryngoscopy, the view will be often grade 3 or 4.⁷⁹ Once manual inline neck stabilization has been applied, the rigid collar should be removed and then reapplied when successful placement of the tracheal tube has been confirmed. Direct laryngoscopy and oral intubation are the quickest method for securing the airway in a patient with potential cervical spine injury.

Many clinicians advocate awake intubation as the safest approach in a patient with a cervical spine injury. It is thought that preservation of muscle tone provides protection, and spinal integrity can be monitored during airway manipulation.⁸⁰ Prolonged hypotension or malposition after intubation is at least as likely to cause neurologic injury as the intubation itself.⁸¹ Awake techniques are particularly favored when the need for intubation is not urgent. Awake intubation is slower than RSI. In the acute trauma patient, it may increase the risk of aspiration and can increase intracranial pressure. Awake techniques, such as blind nasal, blind oral, and fiberoptic intubation, require considerable training and the patient has to be cooperative. Of the awake techniques, fiberoptic intubation is best in the patient with a cervical spine injury. The laryngeal mask airway (LMA), intubating LMA, and ProSeal LMA are useful alternatives if tracheal intubation is not possible. The LMA can be inserted easily in the neutral position and is an option in patients with cervical spine instability.⁸² The standard LMA and intubating LMA cause temporary pressures of greater than 250 cm H₂O against the posterior pharyngeal wall during insertion in cadavers. This pressure is sufficient to induce up to 2 mm of displacement of C3.⁸³ During insertion, the intubating LMA causes some movement of the upper cervical spine, but it is less than that produced during direct laryngoscopy.⁸⁴ Neck stabilization methods make insertion of the LMA and intubating LMA more difficult, although the effect on the intubating LMA is less.⁸⁵

Open Penetrating Chest Wounds and Tension Pneumothorax

An open penetrating chest wound is a challenging injury that is potentially associated with life-threatening airway compromise requiring complicated emergency airway management.⁸⁶ Intubation is challenging because preoxygenation may be less effective than usual; furthermore, if mask ventilation is attempted, air may be forced into the subcutaneous tissue, further distorting the anatomy.⁸⁷ In one study of thoracic trauma, approximately 20% of the patients suffered a pneumothorax and/or hemothorax, injuries were penetrating in approximately 10%, and only approximately 3% required a thoracotomy.⁸⁸ Dyspnea, hemoptysis, subcutaneous emphysema, pneumomediastinum, and pneumothorax are found commonly in patients with open, penetrating chest wounds (secondary to blunt trauma). Such a difficult airway needs to be secured; alternative airway techniques have been recommended as first-line approach, in particular, fiberoptic laryngoscopy. Fiberoptic intubation is the “gold standard” whenever the airway is expected to be difficult; direct airway injury can be diagnosed definitively, and an endotracheal tube can be placed into one of the bronchi if there is a major injury of the opposite side.⁸⁹ The patient should breathe spontaneously for

as long as possible and ideally undergo a gas induction of anesthesia followed by bronchoscopy; a rigid scope may be useful if there is blood or copious secretions in the pharynx.

Traumatic Brain Injury

Outcome of traumatic brain injury depends heavily on initial and admission Glasgow Coma Scale score and age.⁹⁰ Direct trauma-related destruction of brain structures (primary lesion) cannot be saved therapeutically. Secondary brain injury, however, defined as the damage to neurons owing to a systemic physiologic response to the initial injury, can be influenced. Hypotension and hypoxia are major causes of secondary brain injury.⁹¹ One of the first tasks of trauma resuscitation is airway management. Approximately 50% of traumatic brain injury patients are reported to be hypoxic in the field, a finding associated with increased mortality.⁹² Indications for intubation include an inability to maintain and to protect the airway but also may include inadequate ventilation, need to hyperventilate, hemodynamic instability, and need for radiographic imaging and subsequent surgery. The common coexistence of cervical spine injury requires vigilance. Early orotracheal intubation is recommended if the Glasgow Coma Scale score is 8 or less, although retrospective studies revealed increased mortality with field intubation.^{93,94} Intubation usually can be accomplished without sedation and pharmacologic paralysis. Sedation and neuromuscular blockade can be useful in optimizing transport of the head-injured patient, but they interfere with neurologic examination and influence initial evaluation and management.⁹⁵ Hemodynamic stability similarly is ranked as a priority before and after intubation. These priorities should not be compromised in an attempt to prevent an increase in intracranial pressure.

Intracranial hypertension itself is not harmful unless it causes cerebral perfusion pressure to fall below a critical value. Cerebral ischemia leads to neuronal injury and cerebral edema, which further increases intracranial pressure, progressing to irreversible neurologic damage. Proper maintenance of gas exchange is more important than pharmacotherapy aimed at intracranial pressure control. Hyperventilation reduces intracranial pressure by causing cerebral vasoconstriction with a subsequent reduction in cerebral blood flow. Long-term hyperventilation is no longer recommended because outcome is worse than with normocapnia^{96,97}; initial target partial pressure of arterial carbon dioxide (PaCO_2) should be 35 to 40 mm Hg.⁹⁸ Short-term hyperventilation, however, may have a role in reducing intracranial pressure in patients who are deteriorating rapidly before other measures can be instituted.⁹⁹ The lowest level of positive end-expiratory pressure that maintains adequate oxygenation and prevents end-expiratory alveolar collapse should be used.⁹² Hemodynamic stability has particular importance in serious traumatic brain injury patients because outcome is worse with systolic pressures of less than 90 mm Hg¹⁰⁰ and cerebral perfusion pressures of less than 70 mm Hg. Thus, anaesthesia for intubation has to be performed as cautiously as possible maintaining a stable blood pressure.¹⁰¹

Burns

In a burn patient, airway and respiratory complications remain a common cause of morbidity and mortality. Multiple variables have an impact on resuscitation, including delay in its initiation, inhalation injury, and the depth and vapor-transmission characteristics of the wound itself. Inaccurate volume administration causes substantial airway, respiratory, and other morbidity; thus, burn resuscitations must be guided by hourly evaluation of resuscitation end points.¹⁰² Constricting circumferential or near-circumferential torso wounds can reduce chest wall compliance as soft tissues swell beneath the inelastic eschar. Improvement in ventilation is common after escharotomy of the chest and abdomen.

In the initial evaluation and resuscitation phase, the key point of airway management is assessment of airway patency and inhalation injury. The latter results from aspiration of superheated gases, steam, or noxious products of incomplete combustion. Inhalation injury adversely affects gas exchange and hemodynamics. It also profoundly influences mortality from that predicted by age and burn size.¹⁰³ Physical signs of particular importance are singed nasal hair, facial burns, and soot in the mouth and between the teeth. Stridor mandates rapid intubation. In questionable cases, direct laryngoscopy and fiberoptic bronchoscopy can be very helpful; the bronchoscope can serve as a stylet for intubation. Patients at risk for progressive edema should be observed closely and intubated early. After securing the airway, the following require close attention: bronchospasm, small airway obstruction, carbon monoxide poisoning, and respiratory failure. If massive edema is associated with burn resuscitation, reintubation after unplanned extubation can be especially difficult; prevention by endotracheal tube security is the best approach, such as securing the endotracheal tube with an umbilical tie harness. Adjunctive techniques in case of difficult intubation, such as laryngeal mask airway or a needle or open cricothyroidotomy, can be lifesaving.

Drowning

Worldwide, 3.5 deaths per 100,000 population are caused by drowning accidents. Death by submersion is the second most common cause of accidental death in children.¹⁰⁴ Victims of drowning accidents usually show an initial phase of panic and swimming movements. Apnea and breath holding often are followed by swallowing of large amounts of fluid with subsequent vomiting, gasping, and fluid aspiration. Ultimately, severe hypoxia leads to unconsciousness, loss of airway reflexes, and further movement of water into the lungs. Aspirated hypotonic or hypertonic fluids result in bronchospasm, leading to an increase in relative shunt and alveolar edema.¹⁰⁵ These events cause hypoxemia, decreased lung compliance, and increased work of breathing. Up to 70% of drowning victims aspirate foreign material such as mud, algae, and vomitus. Patients usually develop acute lung injury followed by acute respiratory distress syndrome within a very short time. Because hypoxia is the major cause of death, the primary goal is to restore [oxygen](#) delivery. Immediate rescue out of the water is critical. After initial resuscitation, continuing heat loss must be prevented by adequate insulation against the environment.¹⁰⁶ Patients presenting awake or somnolent but with clinical signs of respiratory distress should receive [oxygen](#) at a high concentration using a tight-fitting mask and a reservoir bag. In the case of progressive deterioration of respiratory or neurologic function, RSI has to be performed and ventilation with positive end-expiratory pressure and 100% [oxygen](#) initiated without delay. A nasogastric tube will decompress a full stomach and aid ventilation in these patients. Asystole and

ventricular fibrillation warrant aggressive CPR because the prognosis is dismal; unfortunately, CPR is underused in drowning victims.¹⁰⁷ In an investigation of unwitnessed out-of-hospital cardiac arrest, near-drowning was an independent predictor of survival.¹⁰⁸

Adjuncts for Oxygenation, Ventilation, and Airway Control

Use of Oxygen

During resuscitation, 100% inspired oxygen ($FI_{O_2} = 1$) should be used as soon as possible. Exhaled air contains approximately 17% oxygen and approximately 4% carbon dioxide.^{109,110} Ventilation with such a gas mixture of 17% oxygen and 4% carbon dioxide in animals¹¹¹ and healthy, conscious volunteers resulted in insufficient oxygenation and ventilation.¹¹² Short-term therapy with pure oxygen is beneficial and not toxic. Oxygen toxicity occurs only during prolonged therapy with a high FI_{O_2} . During bag-valve-mask ventilation using room air (21% oxygen) and a normal cardiac output, tidal volumes of 700 to 1000 mL (approximately 8.5 mL/kg) were required to maintain adequate oxygenation.^{113,114}

Careful preoxygenation is essential to increase safety before induction of anesthesia and intubation in emergency settings such as trauma victims. Completely preoxygenated lungs can provide a supply of oxygen for up to 10 minutes by resorption of oxygen out of functional residual capacity.¹¹⁵ This time can be further extended secondary to diffusion respiration.¹¹⁶ Hemoglobin streaming through the pulmonary vessels continuously absorbs oxygen out of the functional residual capacity of the lungs (approximately 200 mL/min), whereas carbon dioxide remains (because of its high solubility) mainly in the bloodstream and only a small amount diffuses into the lungs (approximately 40 mL/min). Because the balance constitutes a negative amount of approximately 160 mL of gas per minute in the lungs, a subatmospheric pressure develops, which sucks gas into the lungs from the upper airways. If oxygen is administered continuously, for example, over a nasopharyngeal tube, partial alveolar oxygen pressure may be maintained at levels sufficient for arterial oxygenation; hypercarbia is generally well tolerated (unless brain injury or pulmonary hypertension).¹¹⁵ Thus, oxygenation of the body had been maintained for up to 1 hour in animal settings and, in clinical settings, apnea times of up to 15 minutes have been reported.^{115,117} Of note, in acute shock states in animal experiments, apnea times were dramatically shorter until oxygen saturation in the blood dropped¹¹⁸; nevertheless, the better preoxygenated are the lungs of a patient, the more reserve is available for apnea time.^{115,118}

Ventilation Devices

Bag-Valve-Mask Ventilation

A simple, portable bag-valve-mask device usually is available in hospital wards and in every emergency medical service. Proper bag-valve-mask ventilation is a fundamental skill of resuscitation and should receive a high priority in training. Unfortunately, these skills are poorly performed even after retraining, and inadvertent hyperventilation is detrimental to morbidity and mortality. With an unprotected airway, the risk of stomach inflation and aspiration is high using standard bag-valve-mask resuscitators. Short inspiratory times, high flow rates, and high airway pressures, caused by squeezing the bag too hard and too fast, are the main causes of stomach inflation. Reduced expiratory times produced by inadvertent hyperventilation stacks air in the lungs, creating decreased cardiac refill and decreased coronary perfusion pressure. A small self-inflatable bag was advantageous over a large one when paramedics undertook bag-valve-mask ventilation with a pediatric versus an adult-sized self-inflatable bag using an in vitro model of ventilation with an unprotected airway. Pulmonary ventilation was similar with both bags, but stomach inflation was much less with the pediatric bag.^{119,120}

Depending on lung compliance, different tidal volumes caused stomach inflation in patients, but most often at a peak inflation pressure of greater than 20 cm H₂O. Data from an in vitro model of an unprotected airway confirmed that longer inflation times produced increased lung ventilation and reduced stomach inflation, especially when intrapulmonary airway resistance was high. Although the bag-valve mask is the simplest and most-cost-effective ventilation device available, its efficacy is diminished by lack of skill of the user, the “incident stress” associated with treating a patient in cardiac arrest, and the subsequent inadvertent hyperventilation that these issues cause. A relatively new bag-valve-mask technology exists that assists rescuers in ventilating correctly by responding to squeeze and release of the bag and controlling the flow of gas from the bag, reducing airway pressure, increasing inspiratory times, and reducing the risk of stomach insufflations even in the hand of less-well-trained rescue personnel.¹²¹

Automatic Transport Ventilators

In prehospital care, mechanically powered devices are as effective as other devices for ventilating intubated patients. In unintubated patients, it depends on the mode of power; that is, in pressure-cycled devices, because these device must achieve a relatively high airway pressure (usually greater than the reported lower esophageal sphincter pressure) to turn off the flow and cycle to an exhalation, they create large amounts of stomach inflation, especially as the airway resistance increases or lung compliance decreases. The guidelines for resuscitation recommend avoiding these devices. On the other hand, tidal volume should be restricted in volume- and time-cycled ventilators. In an in vitro model of an unprotected airway, significantly less stomach inflation was found when applying a tidal volume of approximately 500 mL versus approximately 1000 mL.¹²² Controlling inflation time, flow rate, and flow waveform with a mechanical ventilator may be the best solution to control and limit peak inflation pressure for a given tidal volume, but these variables are not controlled easily during manual ventilation.

Airway Control Devices

Oropharyngeal and Nasopharyngeal Airways

These airway adjuncts can be used in unintubated patients requiring ventilation. The oropharyngeal airway should be inserted only in unconscious patients; otherwise, gagging and vomiting can be evoked. The nasopharyngeal airway is better tolerated in somnolent patients. The devices should be inserted carefully by trained personnel; lubricants ease insertion.

Endotracheal Tube

As early as possible during resuscitation, trained personnel should intubate the trachea. Before commencing intubation, the patient should be preoxygenated either by spontaneously breathing pure oxygen or by controlled bag-valve-mask ventilation with a high inspired concentration of oxygen. Preoxygenation increases the alveolar pressure of oxygen; elimination of nitrogen increases the reservoir of oxygen fivefold and generates a grace period of several minutes before a patient becomes hypoxic. The grace period depends on the alveolar volume, shunts, oxygen consumption, and oxygen-carrying capacity. The interruption of ventilation should be as brief as possible, and adequate ventilation and oxygenation must be provided if more than one intubation attempt is necessary. To facilitate intubation, the sniffing position and hyperextension of the head at the atlantooccipital joint is beneficial. A stylet, which should not extend over the distal end, can be used to provide some stiffness to the tube and makes control during insertion easier. When the cuff of the tracheal tube is inflated and the airway thus secured, cricoid pressure can be removed. Tube placement is confirmed by chest excursion and auscultation; over the epigastrium, no stomach gurgling should be heard during inspiration. These clinical signs are not always reliable¹²³; unrecognized misplacement of the endotracheal tube occurred in up to 25% of intubated patients in the field.^{6,7} Verification by direct visualization of the tube passing through the vocal cords and use of end-tidal carbon dioxide monitoring also are recommended.^{124,125} Clinical assessment and esophageal detector devices may be used but are not reliable in all patients.

Laryngeal Mask Airway

The LMA is now used widely for managing failed intubations or difficult airways.^{126,127} It should be considered first among the alternative airway devices. Placement and use of an LMA are simpler than tracheal intubation because laryngoscopy and visualization of the vocal cords are unnecessary. The LMA is introduced into the hypopharynx, and the cuff seals the larynx after inflation. While the arrow is positioned at the upper esophageal sphincter without sealing it, the distal opening of the tube is just above the glottis, providing ventilation and oxygenation. The main limitation of the LMA is a lack of protection against aspiration,¹²⁸ although regurgitation is less likely than with bag-valve-mask ventilation.¹²⁹

A recently developed ProSeal LMA has an additional lumen to introduce a nasogastric tube to drain the stomach or regurgitated fluids¹³⁰ away from the respiratory tract.¹³¹ The bowl of the mask is deeper than that of a standard LMA, with an additional cuff on the dorsal side. Because of these modifications, the airway sealing pressures achievable with the ProSeal LMA are at least 10 cm H₂O higher than with the standard LMA. As with the standard LMA, the ProSeal LMA can be inserted in the neutral position, making it attractive in

potential cervical spine injury. Another limitation of the LMA is the relatively low leak pressure, 19 to 22 cm H₂O,¹³² which may decrease tidal volume and increase stomach inflation.¹³³

The intubating LMA is a modified conventional LMA; once inserted, it allows passage of a tracheal tube, either blindly or by fiberoptic bronchoscopy. Standard insertion requires a neutral position; manipulation of the head and neck is not needed. In a recent study in Bordeaux, France, “senior emergency physicians” achieved a nearly 100% success rate in advanced out-of-hospital airway management applying the intubating LMA as a first-choice alternative to conventional intubation.¹³⁴ Again, this study indicates that it is not only a question of which airway device is applied but which airway device is applied by which rescuer.

Combitube

The Combitube is essentially a double-lumen tube that is inserted blindly through the mouth and is more likely to pass into the esophagus (approximately 95%) than into the trachea (approximately 5%). The esophageal lumen is closed distally and perforated at the hypopharyngeal level with several small openings; the tracheal lumen is open distally. First, the proximal large pharyngeal balloon is inflated, thereby filling the space between the base of the tongue and the soft palate. Second, the distal cuff is inflated. These cuffs provide a good seal of the hypopharynx from the oropharynx and stability in the trachea or esophagus, respectively. The most common reason for failure to ventilate is placing the Combitube too deeply; the entire perforated pharyngeal section enters the esophagus. Pulling the Combitube back 3 to 4 cm usually resolves the problem. The success of insertion and ventilation with the Combitube is comparable with that of other upper airway devices¹³⁵ and is 100% in an urban environment where trauma patients receive care.¹³⁶ The Combitube, however, is not well tolerated by patients with a persistent, strong gag reflex. It should be exchanged to an alternative airway as early as possible. Advantageously, the Combitube can be a routine device in emergency medicine¹³⁷ and in “cannot ventilate, cannot intubate” situations.¹³⁸ The Combitube has the same limitations as the LMA. Thus, it may be unsuitable in patients with hypopharyngeal pathology or preexisting esophageal pathology, such as a malignancy or esophageal varices.¹³⁹

Laryngeal Tube

The recently introduced single-lumen laryngeal tube can be inserted orally without additional equipment and is effective for ventilating and oxygenating patients who experience a respiratory arrest during induction of anesthesia.¹⁴⁰ Thus, the laryngeal tube may be used as an alternative airway device during routine or emergency airway management. Handling of the laryngeal tube has been simplified by blocking two cuffs with one instead of two catheters; the first secures inflation of the oropharyngeal cuff, and the second secures inflation of the esophageal cuff because of different resistance characteristics of the connected tubing. Given its design, the laryngeal tube may not be the best device for spontaneously breathing patients.¹⁴¹ The blind ending in the esophageal inlet also may cause esophageal rupture in the case of vomiting.¹⁴² Accordingly, the laryngeal tube has been fitted with a second lumen for suctioning and free stomach drainage (laryngeal

tube S) but not for ventilation, as with the Combitube. In contrast to the Combitube, the laryngeal tube S has only one adapter, which may be connected with a ventilation device, whereas the remaining connector can only be connected to a suction adapter. This may achieve additional patient safety; it prevents an inexperienced user from inadvertently attaching a ventilator or bag-valve-mask device to the esophageal tubing, which could result in stomach inflation and subsequent ventilation-related complications.

Videolaryngoscopy

In the past, patients' tracheas were being intubated either by direct laryngoscopy or by awake fiberoptic access in expected difficult airway scenarios. In the last years, different portable laryngoscopes with integrated fiberoptic systems have been developed. These systems enable an indirect view of the glottis from the tip of the laryngoscope, and transmit that picture via a special monitor. Thus, the view of the glottis is often dramatically improved, enabling successful intubation as compared with conventional direct laryngoscopy.^{143,144} It is noteworthy that the view of the glottis via a videolaryngoscope does not automatically signify that intubation will be successful. The human airway is kinked and often it may be necessary to compensate for this through use of a preformed stylet in order to place the tip of the tube into the glottic area during application of videolaryngoscopy.¹⁴⁵ The learning curve for these indirect videolaryngoscopic devices, however, is flat, and rescuers are often able to provide successful intubation after less than a dozen intubation attempts.^{143,144,146,147} Thus, employment of videolaryngoscopy may increase in emergency intubation settings within the next years, even in the field, because of high intubation-success rates.

Tracheostomy and Cricothyroidotomy

When the airway is compromised by trauma, or when massive oropharyngeal or hypopharyngeal pathology is present, emergency access may be possible only through an emergency tracheostomy or cricothyroidotomy. Emergency tracheostomy usually is performed via a vertical incision from the cricoid cartilage down in the direction of the sternal notch. A skilled operator can insert a small, cuffed endotracheal tube rapidly.

Emergency cricothyroidotomy is a valid alternative for a less-skilled operator. The cricothyroid membrane, palpable directly under the skin, is incised in its inferior third to minimize the possibility of bleeding. The cricothyroid membrane is not always easy to appreciate (in obese patients or those with short necks).

Cricothyroidotomy is performed infrequently; in inexperienced hands, success is only 60% to 70%.¹⁴⁸ The main advantage of this technique is the blunt dissection of the subcutaneous tissues all the way to the cricoid membrane. An airway catheter is then introduced over a dilator threaded over the guidewire.¹⁴⁹ This technique allows the ultimate insertion of an airway that is considerably larger than the initial needle or catheter. Its internal diameter often is sufficient to allow ventilation with conventional ventilation devices, suctioning, and spontaneous ventilation.

Needle cricothyroidotomy is another alternative, regardless of whether it is surgical or percutaneous. Needle cricothyroidotomy always requires the use of a jet device to provide ventilation. It is associated with a high

incidence of complications, such as massive subcutaneous emphysema, barotrauma with pneumothorax or tension pneumothorax, and air trapping with severe hemodynamic instability.

Conclusion

The goal of ventilation strategy during resuscitation is to optimize oxygenation and carbon dioxide elimination. This can be achieved in an unprotected airway with techniques such as mouth-to-mouth ventilation, but preferably with bag-valve-mask ventilation. Because ventilation of an unprotected airway may result in stomach inflation and subsequent severe complications, securing the airway with an endotracheal tube is the “gold standard”; however, poorly performed intubation can result in esophageal intubation and disaster. Thus, and importantly, airway-device complications are more related to training than to the devices themselves. Excellent success in emergency ventilation depends on initial training, retraining, and actual frequency of performing a given procedure on the job. Rescue personnel with less experience in advanced airway management should step back and try to avoid anesthesia induction and intubation when patients can be oxygenated by spontaneous breathing. If artificial ventilation is unavoidable, for example, during CPR, less-experienced rescuers can apply extraglottic airway devices or retreat to bag-valve-mask ventilation with high FiO_2 . Success in out-of-hospital airway management is often less a question of the “right-tools” than one of the “the right tools in the right hands at the right time.”¹⁵⁰

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